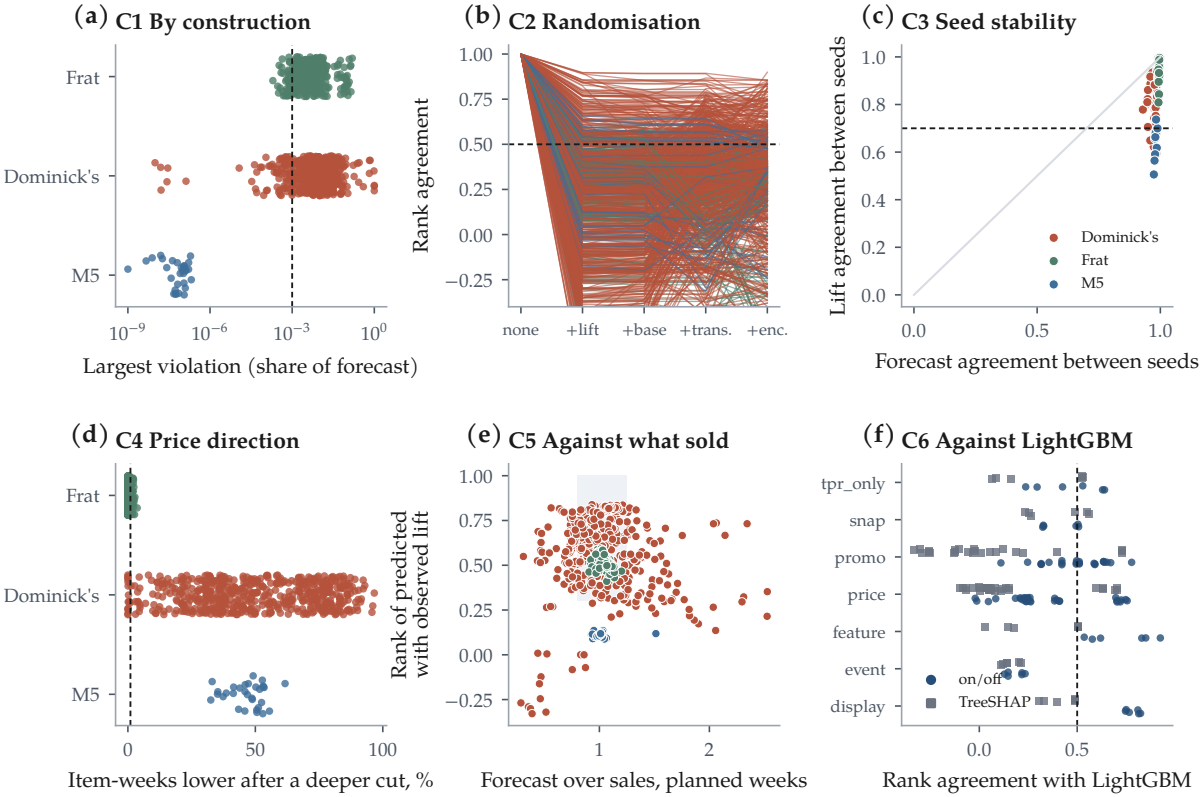

(a) C1 By construction
Frat
Dominick's
M5
10^{-9}
10^{-6}
10^{-3}
10^{0}
Largest violation (share of forecast)
(b) C2 Randomisation
Rank agreement
1.00
0.75
0.50
0.25
0.00
−0.25
none
+lift
+base
+trans.
+enc.
(c) C3 Seed stability
Lift agreement between seeds
0.0
0.2
0.4
0.6
0.8
1.0
Dominick's
Frat
M5
0.0
0.5
1.0
Forecast agreement between seeds
(d) C4 Price direction
Frat
Dominick's
M5
0
50
100
Item-weeks lower after a deeper cut, %
(e) C5 Against what sold
Rank of predicted with observed lift
1.00
0.75
0.50
0.25
0.00
−0.25
1
2
Forecast over sales, planned weeks
(f) C6 Against LightGBM
tpr_only
snap
promo
price
feature
event
display
on/off
TreeSHAP
0.0
0.5
Rank agreement with LightGBM